Tanmay Bhuskute

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SUMMARY

Software Engineer with production experience building and operating large-scale, fault-tolerant backend systems, REST APIs, and cloud infrastructure using Python, Java, Node.js, SQL, AWS, Docker, and Kubernetes, with strengths in cross-functional collaboration, mentorship, and automated testing/CI-CD pipelines. Current research applies LLM-based systems to Android test automation.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go, Ruby.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs, Microservices.

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch.

Developer Tools: Git, GitLab CI/CD, Jenkins, CircleCI, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube, DataDog.

AI & Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments serving production customers.
- Executed **44 production environment upgrades** in 4 months, independently owning diagnosis and resolution of live infrastructure issues across Linux, EC2, and database services to keep systems reliable and secure.
- Designed **Ansible-based Blue-Green deployment** framework cutting upgrade downtime by **50%**; migrated **40+ environments** to Kubernetes via ArgoCD, Helm, and GitLab CI/CD, improving deployment reliability and scalability.
- Served as Quality Coach and SonarQube Specialist, increasing automated test coverage from **35% to 78%** by improving testing and CI/CD quality gates; mentored **5 software engineers** through structured onboarding and knowledge-sharing.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI, PostgreSQL] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.
- Built RESTful API (Python, FastAPI) serving real-time predictions, integrated with PostgreSQL and Redis caching, containerized with Docker for cloud deployment.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).